SIMATS ENGINEERING

ASSESSMENT TOOL 3

Course: Artificial Intelligence | Course Code: CSA17

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A

Dry Run Evaluation

Code of Conduct:

I P Navya (1925260224) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **P Navya**

Question 1 – Breadth-First Search (BFS)

Given graph: A is the root; B and C are its children; D and E are children of B; F and G are children of C.

Assumption: Children are considered from left to right. The queue is FIFO (First In, First Out). The queue column below shows the queue after the visited node is removed and its unvisited children are added.

Iteration	Queue After Iteration	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

Question 1(a). Write the BFS traversal order.

Answer: A → B → C → D → E → F → G. BFS visits all nodes at one level before moving to the next level.

Question 1(b). Show the queue contents after each iteration.

Answer: The queue contents are shown in the table. The queue starts with A. After visiting A, B and C are added. BFS then visits B and C, followed by D, E, F, and G.

Question 2 – Depth-First Search (DFS)

Assumption: DFS visits the left child before the right child. The stack column is shown from bottom to top; the rightmost item is the top of the stack.

Iteration	Stack After Expansion (bottom → top)	Visited Node
1	C, B	A
2	C, E, D	B
3	C, E	D
4	C	E
5	G, F	C
6	G	F
7	Empty	G

Question 2(a). Write the DFS traversal order.

Answer: A → B → D → E → C → F → G. DFS follows one branch as far as possible and backtracks when it reaches a node with no unvisited children.

Question 2(b). Explain the role of the stack in DFS.

Answer: DFS uses a LIFO (Last In, First Out) stack. To visit the left child first, the right child is pushed first and the left child second. The left child is therefore removed from the stack and visited first.

Question 3 – 8-Puzzle Problem Using BFS

Initial state:

1	**3**	**6**
5	□	**2**
4	**7**	**8**

Goal state:

1	**2**	**3**
4	**5**	**6**
7	**8**	□

□ represents the blank (0). For the dry run, legal blank moves are generated in the order Up, Down, Left, Right, and repeated states are not added again.

First five BFS iterations

States are compactly written as three rows separated by slashes. Example: 136/502/478 means rows 1 3 6; 5 0 2; 4 7 8.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	136/502/478	136/502/478	106/532/478 136/572/408 136/052/478 136/520/478
2	106/532/478	106/532/478 136/572/408 136/052/478 136/520/478	136/572/408 136/052/478 136/520/478 016/532/478 160/532/478
3	136/572/408	136/572/408 136/052/478 136/520/478 016/532/478 160/532/478	136/052/478 136/520/478 016/532/478 160/532/478 136/572/048 136/572/480
4	136/052/478	136/052/478 136/520/478 016/532/478 160/532/478 136/572/048 136/572/480	136/520/478 016/532/478 160/532/478 136/572/048 136/572/480 036/152/478

			136/452/078
5	136/520/478	136/520/478 016/532/478 160/532/478 136/572/048 136/572/480 036/152/478 136/452/078	016/532/478 160/532/478 136/572/048 136/572/480 036/152/478 136/452/078 130/526/478 136/528/470

Question 3(a). Which node/state is expanded in each iteration?

Answer: The first five expanded states are: 1) 136/502/478, 2) 106/532/478, 3) 136/572/408, 4) 136/052/478, and 5) 136/520/478.

Question 3(b). How many states are generated in total?

Answer: Using BFS with the stated move order and avoiding duplicate states, 500 unique states (including the initial state) are discovered by the time the goal state is removed from the queue. The goal is reached on expansion 311. Counts can differ if a different move order or duplicate-state convention is used.

Question 3(c). Write the complete solution path from the initial state to the goal state.

Answer: BFS finds a shortest path of 8 moves. The blank moves are: Right → Up → Left → Down → Left → Down → Right → Right. The states along the path are listed below.

Initial state: 136/502/478

Move 1 (Right): 136/520/478

Move 2 (Up): 130/526/478

Move 3 (Left): 103/526/478

Move 4 (Down): 123/506/478

Move 5 (Left): 123/056/478

Move 6 (Down): 123/456/078

Move 7 (Right): 123/456/708

Goal state: 123/456/780

Question 3(d). Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

Answer: Every legal move has the same cost, so the number of moves equals the path cost. BFS explores states in increasing order of depth: first 0 moves, then 1 move, then 2 moves, and so on. Therefore, the first time BFS reaches the goal, no shorter path can exist. This guarantee assumes the goal is reachable and repeated states are handled correctly.

Conclusion

BFS explores a graph level by level using a FIFO queue, while DFS explores one branch deeply using a LIFO stack. For the 8-puzzle, BFS systematically explores states by number of moves and therefore returns a shortest solution when each move has equal cost.